Assignment 5 Part C: Dashboard Screenshots & Questions

```python
st.set_page_config(page_title="KLB Employee Dashboard", layout="wide")

# create cached app engine
@st.cache_resource
def get_engine():
    return create_engine(
        f"postgresql+psycopg2://{DB_USER}:{DB_PASSWORD}@{DB_HOST}:{DB_PORT}/{DB_NAME}"
    )

# query helper
@st.cache_data(ttl=600)
def run_query(sql, params=None):
    with get_engine().connect() as conn:
        return pd.read_sql(text(sql), conn, params=params)

st.title("KLB: Employee Dashboard")
```

Problems 13 | Output | Debug Console | Terminal | Ports | Jupyter | streamlit

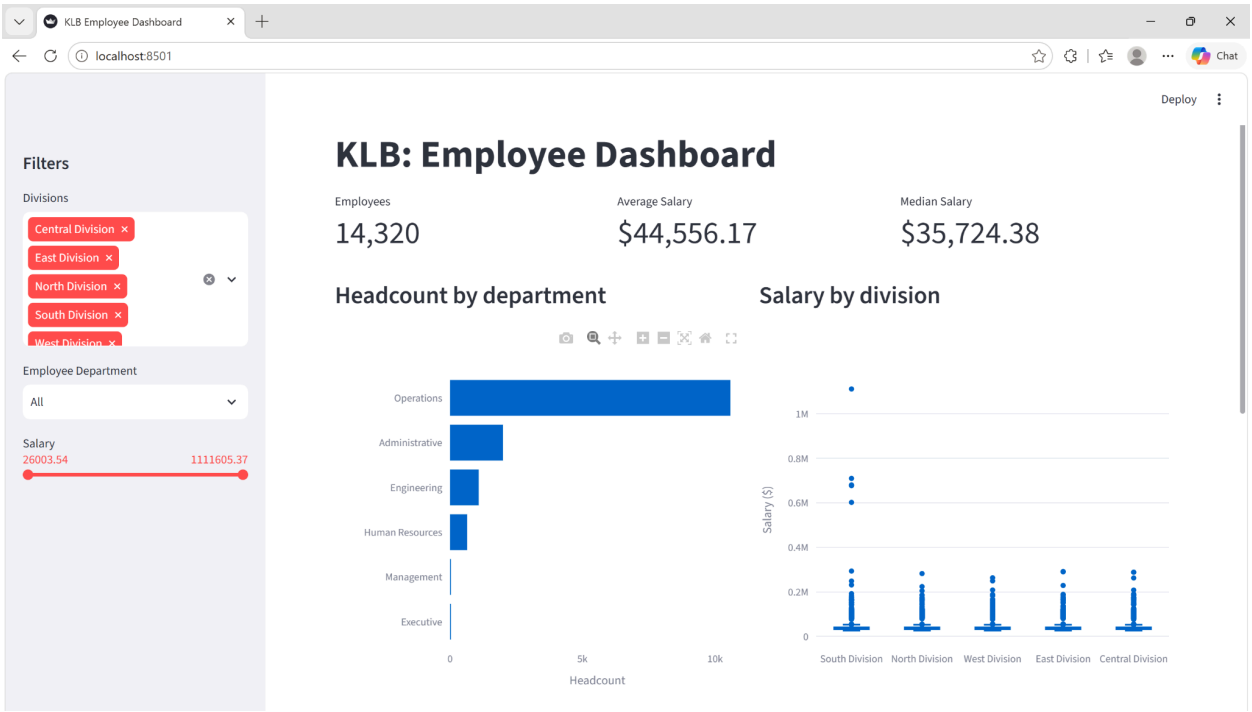
```
PS C:\Users\annaw\OneDrive\Documents\DAEN328\daen328_env\A5> streamlit run app.py
2026-10-29 11:21:59.220 Uvicorn server started on :::8501

  You can now view your Streamlit app in your browser.

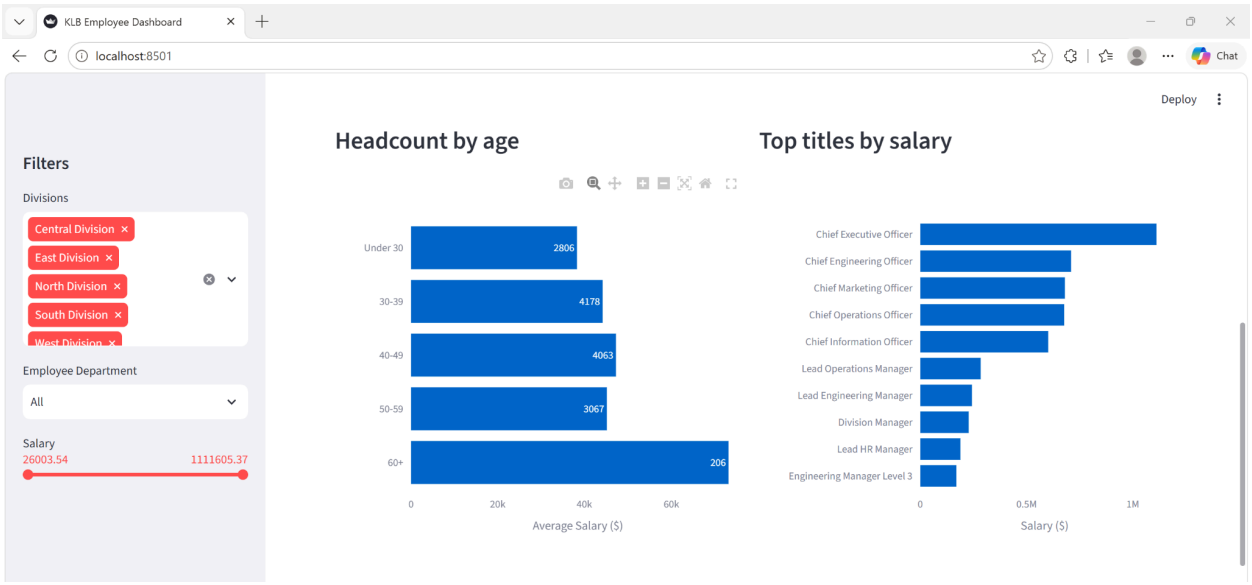
  Local URL: http://localhost:8501
  Network URL: http://10.244.35.169:8501

  Help agents write better Streamlit apps?
  Install the official Streamlit skills by running streamlit skills in your terminal.
```

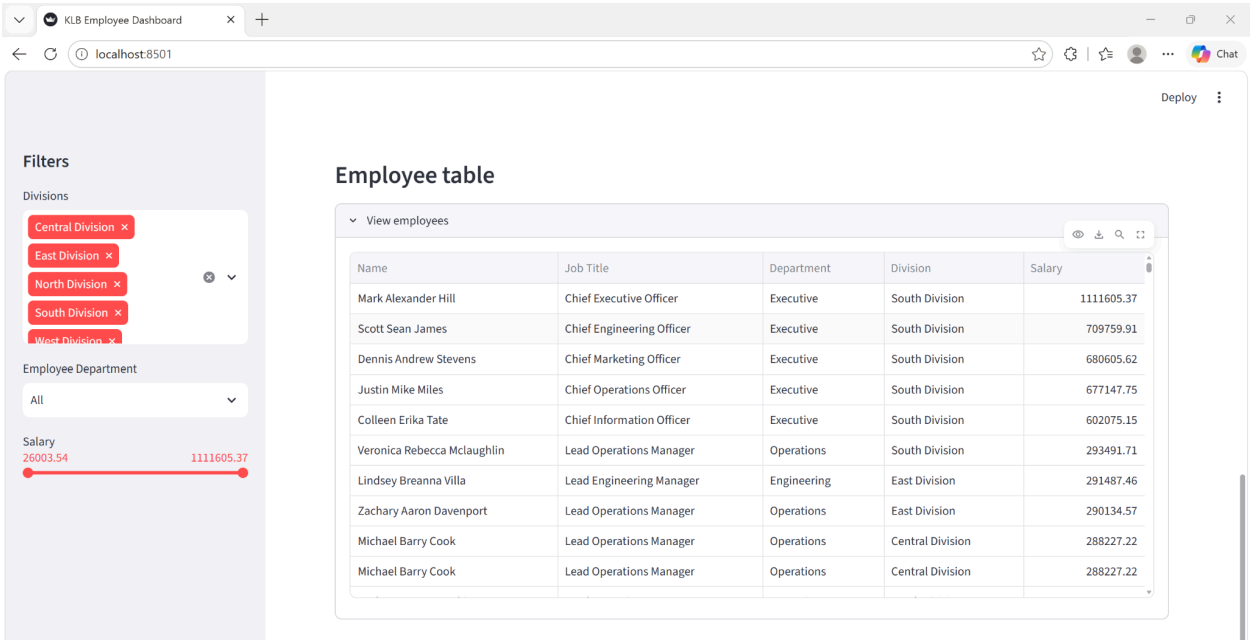
1. Terminal running ‘streamlit run app.py’



2a. Dashboard with no filters: KPIs, headcount by department, and salary by division

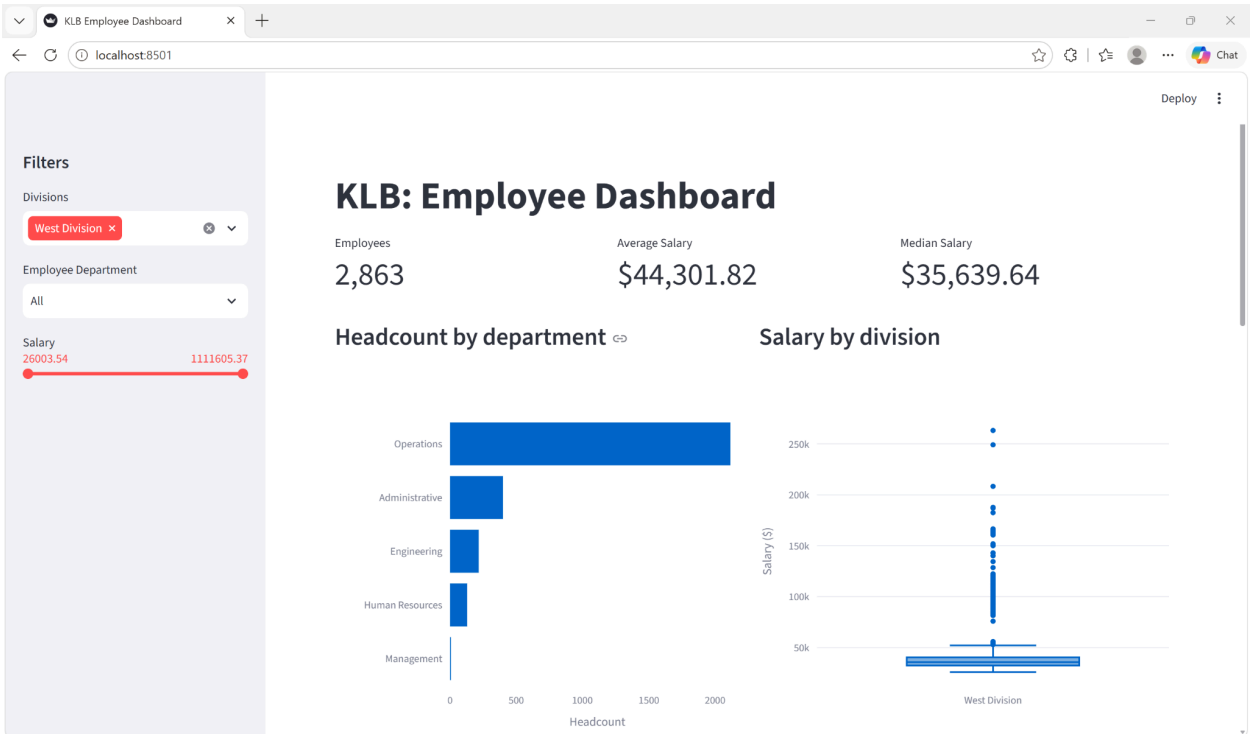


2b. Dashboard with no filters: headcount by age and top titles by salary

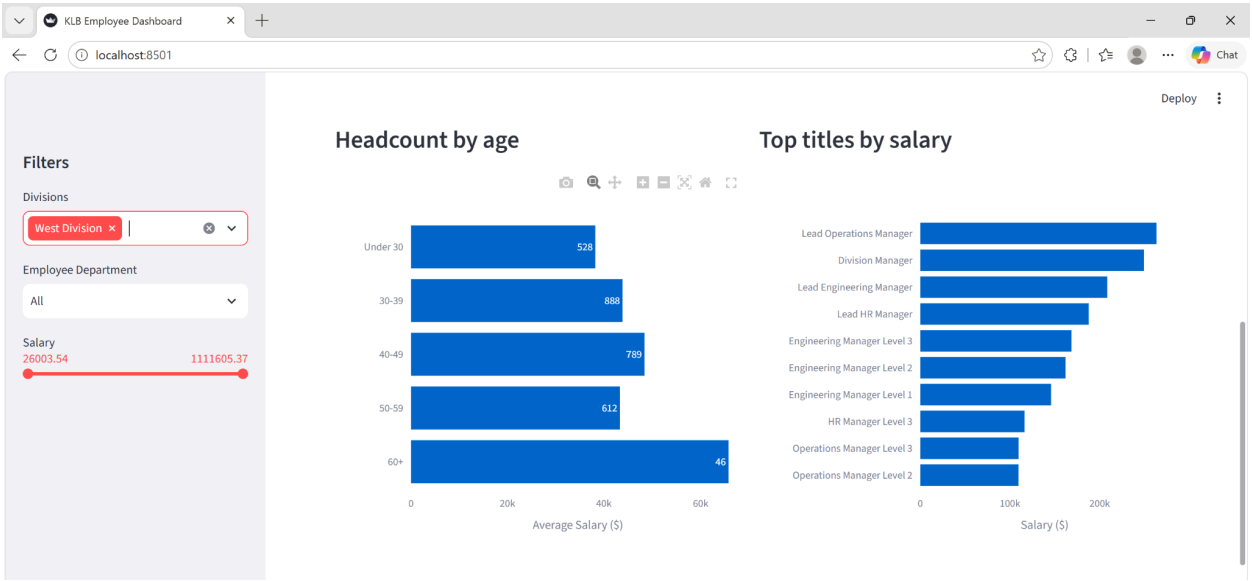


Name	Job Title	Department	Division	Salary
Mark Alexander Hill	Chief Executive Officer	Executive	South Division	1111605.37
Scott Sean James	Chief Engineering Officer	Executive	South Division	709759.91
Dennis Andrew Stevens	Chief Marketing Officer	Executive	South Division	680605.62
Justin Mike Miles	Chief Operations Officer	Executive	South Division	677147.75
Colleen Erika Tate	Chief Information Officer	Executive	South Division	602075.15
Veronica Rebecca Mclaughlin	Lead Operations Manager	Operations	South Division	293491.71
Lindsey Breanna Villa	Lead Engineering Manager	Engineering	East Division	291487.46
Zachary Aaron Davenport	Lead Operations Manager	Operations	East Division	290134.57
Michael Barry Cook	Lead Operations Manager	Operations	Central Division	288227.22
Michael Barry Cook	Lead Operations Manager	Operations	Central Division	288227.22

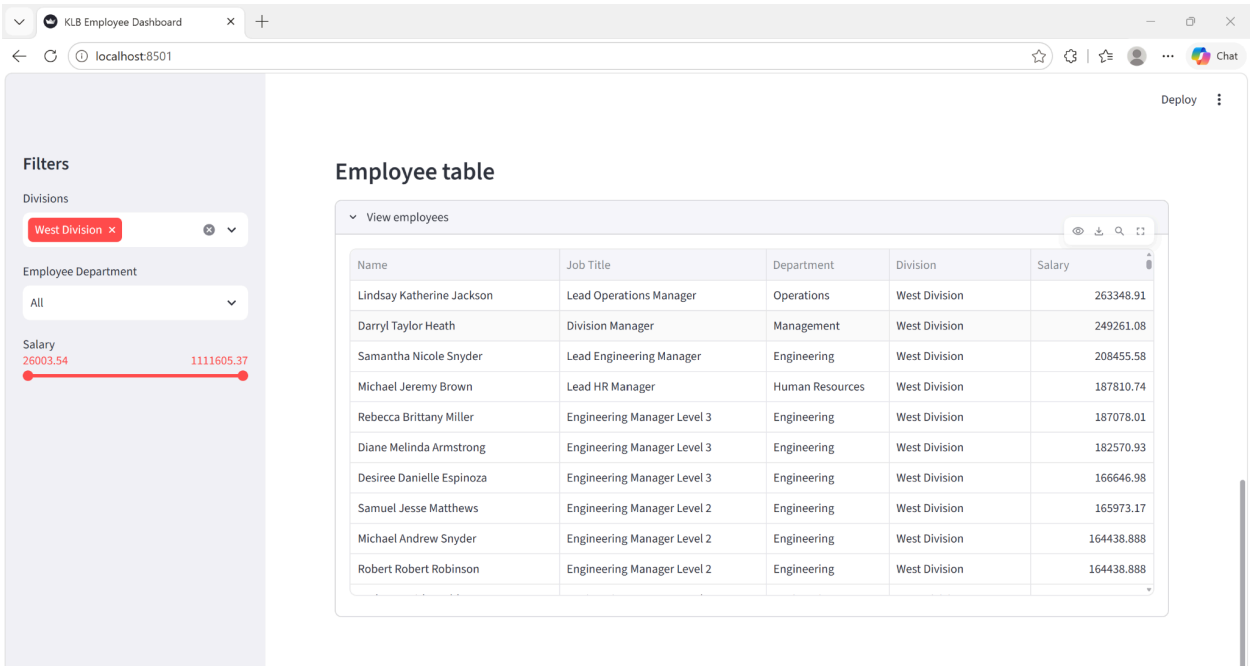
2c. Dashboard with no filters: employee table



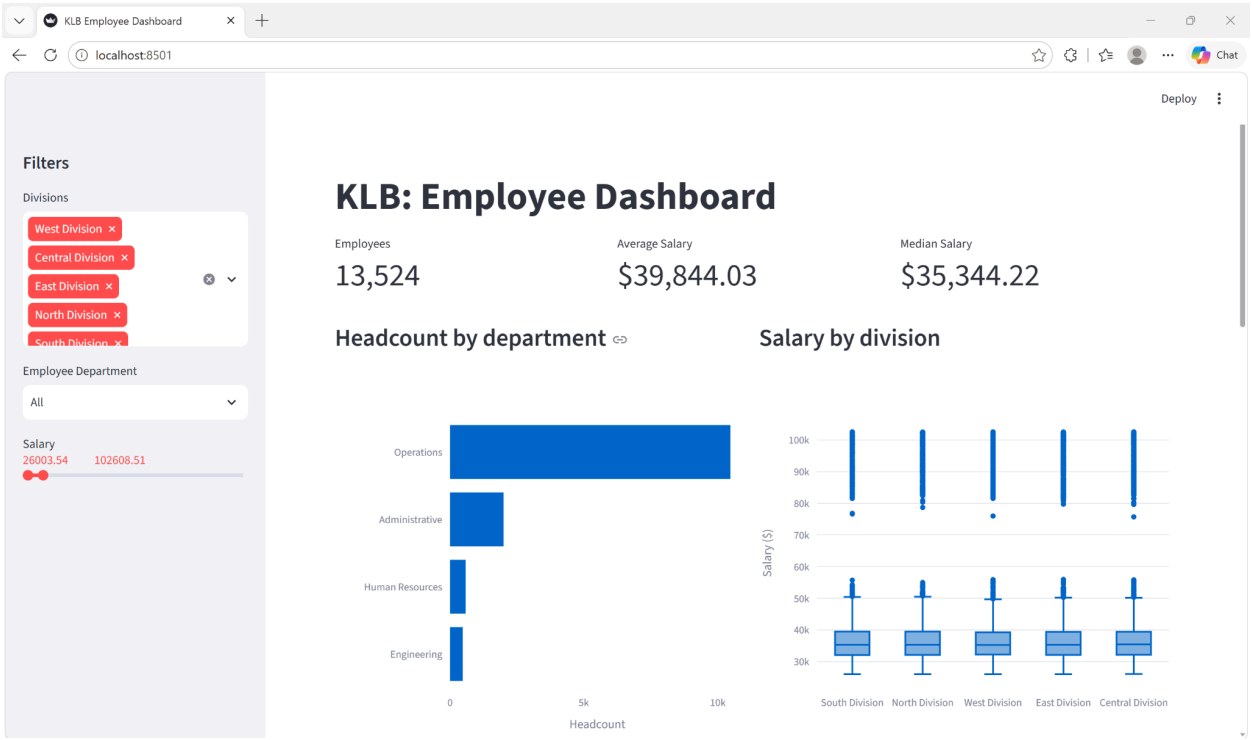
3a. Dashboard with West Division selected: KPIs, headcount by department, and salary by division



3b. Dashboard with West Division selected: headcount by age and top titles by salary



3c. Dashboard with West Division selected: employee table



4. Box plot filtered by salary for question 2

1. *Select only the West Division and report the three KPIs. Why are the average and the median salary so far apart? (1 to 2 sentences)*

   In the West Division, there are 2,863 employees with an average salary of $44,301.82 and a median salary of $35,639.64 (Screenshot 3a). The large difference between the average and median salary is due to some outlier salaries ($150,000+) that bring the average up considerably.

2. *With no filters, the salary box plot is hard to read. Explain what causes this and use your filters to make the boxes readable. Include a screenshot of the readable version and say which filter settings you used.*

   The box plot is difficult to read without filters because there are some significant outliers that cause the boxes to get squished in order to fit all the data points in the limited y-axis space. To address this problem, I adjusted the salary filter to only include salaries in the range of $26,003.54 (minimum salary at company) to $102,608.51, the results of which can be seen in Screenshot 4. If this was still not readable enough, the top of the salary range could be decreased to approximately $60,000.

3. *Which age band has the highest average salary for the whole company? Now select department Operations. Does the band still lead, and how many employees is the average based on? What does this tell you about trusting an average? (2 to 3 sentences)*

   The 60+ age band has the highest average salary for the whole company as well as within the Operations department. However, the average salary within the Operations department for this age group is based on twelve employees, so it would be very easy for one or two large salaries to bring the whole group's average up significantly. This is why averages may not always be trustworthy, especially in small datasets with outliers.